

Space360

Comprehensive Master Roadmap

Version: **v1.0**

Created: **August 21, 2026**

Purpose: AI assistant handoff reference — self-contained context for any new chat session

Maintained at: F:\Space360\docs\SPACE360_MASTER_ROADMAP_v1.0.md

How to use: Share this document at the start of any new AI assistant chat. The assistant reads it to understand full project context, then waits for your first instruction.

1. Project Overview

Space360 is a 360° field inspection platform for construction and site management.

Property	Value
Type	Web app + Mobile app + Standalone modules
Domain	Construction / Site inspection
Primary users	Admin, Manager, Contractor
GitHub	https://github.com/Sriganeshbabu1126/Space360
Project root	F:\Space360\
Docs folder	F:\Space360\docs\

2. Technology Stack

Backend

Component	Technology
Language	Python 3.13
Framework	FastAPI + Uvicorn
ORM	SQLAlchemy + Alembic (migrations)
Database	PostgreSQL 16
Storage	Google Cloud Storage (GCS)
Auth	Firebase Authentication
Task queue	Celery + Redis (added Stage 2)
Search	PostgreSQL GIN + tsvector full-text search

Frontend (Web)

Component	Technology
Framework	React 18 + TypeScript
Styling	Tailwind CSS v3.4.1
Routing	React Router v7
Charts	recharts
Icons	lucide-react
360° viewer	Pannellum.js
Markup	Canvas API
Notifications	react-hot-toast

Mobile App (Stage 3 — in progress)

Component	Technology
Framework	React Native + Expo (managed workflow)

Component	Technology
Platforms	iOS + Android
Navigation	React Navigation v7 (native stack + bottom tabs)
State	Redux Toolkit + redux-persist
Auth	@react-native-firebase/auth
HTTP	Axios
Storage	AsyncStorage
Location	expo-location
Camera	expo-camera

Infrastructure

Component	Value
Backend hosting	Google Cloud Run
Database hosting	Google Cloud SQL (PostgreSQL)
File storage	Google Cloud Storage
GCS bucket	360-field-check-media-sgb
Firebase project	field-check-72967
GCP region	asia-southeast1
Service account	space360-backend@field-check-72967.iam.gserviceaccount.com
CI/CD	GitHub Actions

Stage 3 Additions (planned)

PostGIS · Three.js · Deck.gl · Cloud Tasks + Redis · FFmpeg / OpenCV (Insta360 module)

3. Repository & Local Development

Item	Value
Project root	F:\Space360\
Backend	F:\Space360\backend\
Frontend	F:\Space360\frontend\
Mobile app	F:\Space360\mobile\
Insta360 module	F:\Space360\modules\insta360\
Docs	F:\Space360\docs\
Start script	F:\Space360\START-SPACE360.bat
Backend port	8000
Frontend port	3000

4. User Roles

Role	Permissions
Admin	Full access — all sites, all users, all data
Manager	Site-scoped — manage issues, contractors, reports
Contractor	Restricted — assigned sites only, limited field visibility

5. Development Workflow

Role	Responsibility
AG (Antigravity IDE)	Autonomous developer — executes prompts, writes all code
Human (You)	Strategic decisions, prompt authoring, testing, review
AI Assistant (Claude)	Consultant — drafts AG prompts, reviews plans, answers questions

Process: Plan → draft AG prompt → human sends to AG → AG implements → human tests → review → next prompt

Principles: Quality over speed · One feature complete before next · Stage 1 always backward compatible · Reset AG context explicitly when it drifts

6. Overall Progress

Stage	Name	Status	Completion
Stage 1	Core Platform	■ Complete	100%
Stage 2	Advanced Features	■ Complete	100%
Stage 3	Path Tracking + Mobile + Video	■ In Progress	~10%
Insta360 Module	Standalone video handling	■ Planning	0%

7. Stage 1 — Core Platform ■ COMPLETE

Do not revisit or modify Stage 1 features. All are production-ready. Backward compatibility must be preserved through all future changes.

Features shipped: Issue tracking · Evidence photo upload · Markup/annotation tools · Push notifications · Dashboard · Bulk actions · Mobile-responsive web UI · Multi-role access control (Admin/Manager/Contractor) · Firebase Authentication · GCS file upload/download · 2000+ lines of stable tested code · Ready for Cloud Run deployment

8. Stage 2 — Advanced Features ■ COMPLETE

Feature	ID	Status	Notes
Project Hub	#76	■ Complete	Multi-project management
Multi-Site Contractors	#77	■ Complete	Contractors across sites
Frame Sequence Capture	#78	■ Complete	Sequential 360° capture
Advanced Filtering & Search	#79	■ Complete	B-Tree + GIN + tsvector
Column Sort Fix	#79.1	■ Complete	Tri-state sort, Issues page
Data Export	#80	■ Complete	CSV/Excel/PDF, streaming + async

Key Implementation Details

#79 Filtering: Alembic migration — B-Tree indexes on issues/comments/assignments; GIN indexes with tsvector for full-text search. IssueFilterQuery service with dynamic criteria and contractor-scoped /search endpoint. AdvancedFilterPanel + FilterPresetManager (localStorage).

#79.1 Sort: Tri-state column sort (asc → desc → reset). State: sortColumn + sortDirection. Params: ?sort_by=created_at&sort_dir=asc. Sortable: Title, Status, Priority, Created Date, Updated Date, Assignee.

#80 Export: export_service.py generates CSV/Excel/PDF. Role-based fields. ≤500 issues = streaming download; >500 = Celery + Redis async job → GCS → signed URL (1hr). New endpoints: POST /issues/export + GET /issues/export/{job_id}. GCS path: 360-field-check-media-sgb/exports/{user_id}/{job_id}.{ext}. Celery worker (Windows): celery -A app.celery_app worker --loglevel=info -P threads

9. Stage 3 — Path Tracking + Mobile + Video ■ IN PROGRESS

9.1 Feature Tracker

Feature	ID	Status	Blocking?
Mobile App Foundation	#81	■ In Progress (AG working)	No
Path Tracking (GPS/IMU + PostGIS)	#82	■ Planned	No
Path Navigation (Three.js overlay)	#83	■ Planned	Needs #82
Video Capture (Insta360 X4)	#84	■■ On Hold	Yes — 8 specs unanswered
Mobile: Issue Viewer	#85	■ Planned	Needs #81
Mobile: Path Capture	#86	■ Planned	Needs #81 + #82

9.2 Feature #81 — Mobile App Foundation (In Progress)

Location: F:\Space360\mobile\ · Framework: Expo managed, TypeScript strict · Bottom tabs: Dashboard, Issues, Capture (placeholder), Profile · Firebase Auth (email/password) persisted via redux-persist + AsyncStorage · Axios client with token interceptor + auto-logout on 401 · Brand color: #1D9E75 (teal)

Success criteria: App launches iOS + Android · Login/logout with existing Firebase accounts · Session persists across restarts · No TypeScript errors

9.3 Feature #82 — Path Tracking (Planned)

GPS + IMU capture during walkthroughs · PostGIS on Cloud SQL · Schema: paths table (id, site_id, user_id, started_at, ended_at, geometry) + path_points table (id, path_id, lat, lng, altitude, heading, timestamp) · Accuracy: ±5m · Duration: 5–30 min · Dependency: expo-location (already installed in #81)

9.4 Video Capture — ON HOLD (8 unanswered spec questions)

#	Question	Status
1	Video resolution: 1080p / 4K / 8K?	■ Unanswered
2	Codec: H.264 / VP9 / AV1?	■ Unanswered
3	Quality: CRF value (28 recommended)?	■ Unanswered
4	Path tracking: GPS only or GPS+IMU?	■ Unanswered
5	Accuracy: ±5m acceptable?	■ Unanswered
6	Duration: 5–30 min typical?	■ Unanswered
7	Mobile scope: iOS / Android / both?	■ Unanswered
8	Hardware: Insta360 / LG 360 / native / all?	■ Unanswered

10. Insta360 X4 Video Handling Module ■ PLANNING

Status: Planning — no code yet · **Location:** F:\Space360\modules\insta360\ · **Chat:** Separate Claude chat dedicated to this module · **Approach:** Standalone — built and tested independently before integration

10.1 Planned Scope

Responsibility	Detail
Video ingestion	From Insta360 X4 via USB / SD card / wireless
Format handling	.insv dual-lens files → stitched equirectangular MP4
Metadata extraction	GPS, IMU, heading, timestamp, resolution, bitrate
Quality validation	Corruption detection, completeness check, stitch quality
Format standardization	Normalized video + sidecar JSON for Space360
Storage	Upload to GCS: 360-field-check-media-sgb/videos/
Integration interface	Clean API or CLI that Space360 calls

10.2 Planned Tech Stack

Component	Technology
Language	Python
Video processing	FFmpeg + OpenCV
Interface	FastAPI or CLI (TBD)
Storage	Google Cloud Storage SDK
Initial environment	Local Windows (F:\Space360\modules\insta360\)
Future	Containerized for Cloud Run

11. Key Architectural Decisions

Decision	Choice	Reason
Mobile framework	React Native + Expo	Cross-platform, reuses JS/TS skills
Mobile state	Redux Toolkit + redux-persist	Offline-first, consistent patterns
Full-text search	GIN + tsvector (PostgreSQL)	Native, fast, no extra service
Contractor search scope	Site-restricted	Security requirement
Export async threshold	500 issues	Balances UX vs server load
Export storage	GCS (existing bucket)	Reuse infrastructure
Filter presets	localStorage	Simple, no DB changes
Video processing	Standalone module	Isolates complexity, parallel workflow
Path tracking accuracy	±5m acceptable	Sufficient for site walkthrough
Video specs	ON HOLD	Team not ready — 8 questions pending
Timeline	Flexible	Quality over speed

12. Known Issues & Gotchas

#	Issue	Status
1	Celery + Redis must be running locally for large export dev (#80)	Active
2	Video capture Stage 3 Phase 2 blocked until 8 spec questions answered	Active
3	Stage 1 must remain 100% backward compatible — permanent constraint	Permanent
4	AG can lose context in long sessions — send explicit state reset messages when drifting	Ongoing
5	google-services.json (Android) + GoogleService-Info.plist (iOS) must be placed manually for Firebase	Pending #81

13. Success Criteria

Stage 2 (Complete ■): Advanced filtering + search · Column sort + filters working together · CSV/Excel/PDF export for small and large datasets · All Stage 1 features backward compatible

Stage 3: Mobile login + navigation + session persistence on iOS + Android · Path tracking GPS capture + PostGIS · Three.js path overlay on 360° viewer · Mobile issue viewer for contractors · Test coverage >80%, critical bugs 0

Insta360 Module: .insv → equirectangular MP4 pipeline end-to-end · Metadata sidecar JSON correct · GCS upload working · Integration interface documented and tested

14. Document History

Version	Date	Author	Changes
v1.0	Aug 21, 2026	Claude (AI consultant)	Initial creation — full project handoff document

This document is the authoritative reference for Space360 development. Update it when stages complete, decisions change, or new modules are added.